Response letter for paper *TNSE-2026-02-0321*:

Client-Driven Performance Model of Hyperledger Fabric Blockchain via Phase Decomposition

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We deeply appreciate the constructive comments from the editor and the reviewers, which helped us improve our manuscript's technical accuracy and presentation. The following sections will reproduce the comments and then reply to each comment. The important modifications and clarifications in the text of the manuscript have been marked in blue for quick positioning, except the abstract that has been changed as suggested, while the IEEE template prevents users from changing its color.

1 Responses to Associate Editor

1.1. This paper proposes a phase decomposition-based performance modeling method for Hyperledger Fabric. It breaks down Fabric's transaction processing flow into three distinct phases and conducts separate performance analysis and modeling for each. The authors first design a client-driven measurement framework, and introduce a stochastic computation model and an alpha-beta communication model to analyze performance bottlenecks. Additionally, an optimization strategy is proposed to address the waiting time issue in the order phase. The paper validates the model through experiments conducted in two differently configured clusters, demonstrating its effectiveness. The author should include specific performance improvement values at the end of the abstract to clearly demonstrate the technical advancement of the model/method proposed in this paper.

Thanks for the comment. This is the response. Here are the experiment results. Here are the experiment results. TODO: Specific improvement value.

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1.2. Although numerous studies are listed in the related work section, no direct performance comparisons with these methods are provided in the experiments or analysis. Incorporate performance comparisons with representative methods (e.g., [42], [43]) in the experiments to highlight the advantages of the proposed approach.

TO DO: Here we need to new the plot figures. Here we need to plot the new figures. Here we need to plot the new figures.

2 Responses to Reviewer 1

2.1. The authors sufficiently addressed my previous comments. I therefore advise to accept the paper for publication.

Thanks for the positive comment and good suggestion.

3 Responses to Reviewer 2

3.1. I appreciate the authors for the improvements they made to the prior submission. The quality of writing is satisfactory and the paper is now easy to understand.

Thanks for the positive comment.

3.2. In conclusion, can you please add the future research direction based on the results of this manuscript? can you please explain what issues remain still open after publishing this paper? what are the limitations of this work? and how future works can address them?

Thanks for the positive comment and good suggestion. We add a discussion about one major limitation to the conclusion section in the revised manuscript, and propose a possible future direction to overcome this limitation in the conclusion.

4 Responses to Reviewer 3

4.1. The authors propose that the optimal strategy for a Bitcoin mining pool is to either increase it's mining hash rate or keep it unchanged. However, in real world, we see that the mining hash rate of a pool can, in fact, drop significantly (e.g., see https://www.theblock.co/data/on-chain-metrics/bitcoin/bitcoins-daily-real-time-hashrate-by-pool). Since participants join mining pools voluntarily, they can leave the pool whenever they want thus reducing the overall hash rate of the pool. Why should propositions 4.5 and 4.6 then hold true?

Thanks for the question. Sorry that our previous presentation may create some confusion about the modeling of mining pool behaviours. In Section IV.B.3) "The Overall Computing Power of the Bitcoin Network", we try to understand the evolution of mining pools' computing power from the perspective of game theory. We believe that a mining pool's overall behavior is affected by the Bitcoin price as well as the cost of electricity power and the behavior of other competitors. To this end, we have developed four Propositions: 4.5, 4.6, 4.7, 4.8, which are corresponding to four different scenarios. We have carefully considered the existence of new mining pools and the strategy of existing mining pools, where we assume that all pools (and miners) in the mining game are rational to maximize their mining payoffs. Under some scenarios, the best strategy of a mining pool would be increasing the hash rate; but under some other scenarios, the best strategy of a mining pool would be decreasing the hash rate. The mathematical proofs of propositions 4.5 and 4.6 can be found in Appendix C[1].

Second, thank you for your suggested data at `https://www.theblock.co/data/on-chain-metrics/bitcoin/bitcoins-daily-real-time-hashrate-by-pool`, which is very helpful. Based

[1] Appendix is also available at: https://github.com/Canhui/AppendixBTC

on this information, we analyzed the events of hash rate dropping of mining pools up to Aug 31, 2022 in the subsection C of Appendix G[1].

The results show that most events of hash rate dropping (e.g., May 18, 2020, Jan 10, 2020) are due to low mining revenue per computing power. That is, most of the trace data validate our model (see Theorem 4.9 in the manuscript) and result (see Fig. 5 in the manuscript and Appendix G[1]). Nevertheless, we still found two events of hash rate dropping (i.e., Apr 16, 2021, and Jun 29, 2021) in which mining activities were profitable but our current model cannot explain. Specifically, for the former event of the hash rate dropping on Apr 16, 2021, we believe it is related to the unexpected event of Xinjiang Grid Blackouts[2]. And this unexpected event resulted in a significant dropping in hash rate of several major Bitcoin mining pools such as AntPool and BTC.com[3]. For the latter event of the hash rate dropping on Jun 29, 2021, we believe it is related to a government ban in China. As reported by mainstream media, the government and regulators of China moved to effectively shut down all crypto mining operations across the country since May 2021 for the purpose of mitigating financial risks[4], [5]. And this government-mandated ban in China resulted in a 38% drop in the global hash rate by July 2021[6]. In summary, our model well validates most events of hash rate dropping of mining pools. Though certain non-economic behaviors (e.g., unexpected Grid Blackouts, government regulation) exist, the effect of rational strategies could be evident in the long term, e.g., the ratio of a pool's daily mining revenue to the pool's daily computing power in Fig. 5.

We have included the above discussion in Appendix G.

4.2. A major unsubstantiated assumption in this work is that mining pools cash out their Bitcoin rewards immediately, i.e., pools convert their Bitcoins to USD (or other currencies) and do not hold them for later. This forms the basis of TABLE VI and some discussion on pg ix. Can the authors concretely support this?

Thanks for the comment and good suggestion. One limitation of this paper is the assumption of the immediate Bitcoin reward. Predicting the BTC's future price is challenging, and our current work simply considers that mining pools use the current market price of BTC to determine the profitability of mining. This may bring some attacks like withholding the BTC price, or concerting BTC to some other currencies. For the latter case, using the *ixCrypto index* could be a promising scheme in our future work. Finally, we summarize these discussions and add them to the conclusion section of this revised manuscript.

[2]https://news.bitcoin.com/bitcoin-hashrate-drops-xinjiang-blackouts-blamed-btc-price-slides/
[3]https://www.nasdaq.com/articles/bitcoin-mining-hash-rate-drops-as-blackouts-instituted-in-china-2021-04-16
[4]https://www.voanews.com/a/east-asia-pacific_voa-news-china_why-china-cracking-down-bitcoin/6207912.html
[5]https://www.jbs.cam.ac.uk/insight/2021/geographic-shift/
[6]https://www.reuters.com/world/china/china-central-bank-vows-crackdown-cryptocurrency-trading-2021-09-24/

Table 1: Events of Hash Rate Dropping of Mining Pools up to Aug 31, 2022 (Remark: c=1.2 USD/(EH/Day)=0.1037 USD/(TH/s), also see Fig. 5 in the manuscript and Appendix G[1])

Day	Mining Pool	Hash Rate (EH/s)	Hash Rate Drops Significantly?	Mining Revenue over Computing Power (USD/(TH/s))	Profitability	Pure Conditional Strategy
Mar 18, 2020	AntPool	10.07	Y	0.08 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	F2Pool	16.68	Y	0.08 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	BTC.com	12.57	Y	0.08 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	ViaBTC	7.09	Y	0.08 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
May 20, 2020	AntPool	12.06	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	F2Pool	16.08	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	BTC.com	12.65	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	ViaBTC	6.01	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
Jun 10, 2020	AntPool	12.19	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	F2Pool	18.42	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	BTC.com	14.33	N	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	ViaBTC	6.11	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
Oct 29, 2020	AntPool	12.89	Y	0.1 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	F2Pool	17.77	Y	0.1 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	BTC.com	14.28	Y	0.1 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	ViaBTC	9.71	Y	0.1 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
Apr 16, 2021	AntPool	19.45	Y	0.4 ($> c$)	Profitable	-
	F2Pool	24.07	Y	0.4 ($> c$)	Profitable	-
	BTC.com	16.14	Y	0.4 ($> c$)	Profitable	-
	ViaBTC	14.66	Y	0.4 ($> c$)	Profitable	-
Jun 29, 2021	AntPool	10.87	Y	0.23 ($> c$)	Profitable	-
	F2Pool	10.32	Y	0.23 ($> c$)	Profitable	-
	BTC.com	9.93	Y	0.23 ($> c$)	Profitable	-
	ViaBTC	19.64	Y	0.23 ($> c$)	Profitable	-
Jun 23, 2022	AntPool	30.39	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	F2Pool	10.74	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	BTC.com	10.17	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	ViaBTC	19.56	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
Jul 11, 2022	AntPool	31.76	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	F2Pool	29.8	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 1)
	BTC.com	8.57	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)
	ViaBTC	19.3	Y	0.09 ($\leq c$)	**Non-Profitable**	Proposition 4.8 (Case 2)